



# eSim Semester Long Internship Autumn 2026

## Project Report On Smart 5V Power Supply PCB Design using eSim

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## Acknowledgement

I express my sincere gratitude to the FOSSEE eSim Team at the Indian Institute of Technology Bombay for providing the opportunity to undertake Screening Task 7 - PCB Design using eSim as part of the eSim Semester Long Internship - Autumn 2026. The task offered a practical platform to understand the complete workflow involved in converting an electronic circuit idea into a manufacturable printed circuit board.

I am thankful for the eSim resources, task instructions, and open-source design environment made available for the screening activity. These resources supported the systematic development of the Smart 5V Power Supply, including schematic capture, footprint mapping, board generation, component placement, routing, design-rule verification, three-dimensional inspection, and fabrication-output generation.

I also acknowledge the academic support of the Department of Electronics Engineering, Yeshwantrao Chavan College of Engineering, Nagpur. The project strengthened my understanding of regulated power supplies, PCB design methodology, manufacturing files, and the importance of verification before fabrication.

The experience gained through this work has improved my confidence in using eSim and the KiCad-based PCB design environment for practical electronic hardware development.

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# Chapter 1

## Introduction

The aim of the project is to design and document a two-layer PCB for a regulated 12 V to 5 V DC power supply using eSim and its KiCad-based PCB environment. The scope covers schematic creation, component-footprint mapping, board outline definition, component placement, copper routing, DRC verification, 3D visualization, and generation and inspection of Gerber and drill files. Physical fabrication and load-testing of an assembled board are outside the scope of this screening submission.

### 1.3 Aim and Scope of the Project

For an educational PCB project, the LM7805 provides a straightforward way to study the relationship between schematic design and physical layout. The input and output bypass capacitors demonstrate the importance of local decoupling, while the LED branch provides a simple visual indication that the output rail is energized.

A stable 5 V rail is commonly required by digital logic, microcontroller boards, sensors, interface circuits, and laboratory prototypes. A regulated supply helps keep the output voltage substantially constant despite normal changes in the input source and connected load, provided that the regulator is operated within its electrical and thermal limits.

### 1.2 Need for a Regulated 5 V Supply

The Smart 5V Power Supply was selected as a compact but complete PCB-design exercise. It uses a widely used LM7805 linear regulator to obtain a nominal +5 V DC output from a +12 V DC source. The circuit also includes reverse-polarity input protection, local filtering capacitors, an LED indication branch, and dedicated input, output, and ground connectors.

Printed circuit board design is an important stage in electronic product development because it converts a circuit schematic into a physical interconnection platform suitable for assembly and fabrication. In the eSim workflow, schematic capture is followed by footprint assignment, board generation, placement, routing, design-rule verification, three-dimensional inspection, and generation of manufacturing outputs.

### 1.1 Background

This project presents the design of a simple regulated DC power supply PCB developed using the eSim/KiCad-based PCB design environment. The circuit accepts a nominal +12 V DC input and uses an LM7805 linear voltage regulator to provide a regulated +5 V output. A 1N4007 diode is included at the input, capacitors are used for filtering and regulator stability, and an LED with a 330 ohm series resistor provides a visual power indication.

## Chapter 2

### Project Objectives

A further objective was to maintain a complete and reproducible project folder so that the design can be opened and evaluated in the eSim/KiCad environment without depending only on screenshots. This supports both technical review and future modification of the design.

The project was undertaken with the following engineering objectives: to create a clear LM7805-based power-supply schematic; to select footprints that correspond to the intended through-hole components; to organize the PCB so that the power path is easy to follow; to use both front and back copper layers in accordance with the Task 7 requirement; to route all electrical nets without unconnected items; to verify the final board using DRC; and to produce fabrication-ready Gerber and drill outputs for review.

The objective is to demonstrate the complete PCB design workflow required in Task 7: schematic capture, component footprint assignment, PCB generation, component placement, routing, design-rule verification, 3D visualization, and generation/verification of manufacturing files.

## Chapter 3

### Circuit Architecture and Working Principle

The approximate power dissipated by a linear regulator is related to the voltage dropped across the regulator and the load current,  $P = (V_{in} - V_{out}) \times I_{out}$ . Therefore, although the PCB demonstrates the complete design flow, practical high-current use would require attention to regulator temperature, heatsinking, and the allowable current of the chosen footprints and tracks.

Capacitor C2 (0.1 uF) is connected between the +5 V rail and ground near the regulator output. The LED D2 and resistor R1 (330 ohm) form an output-status indicator branch. The resistor limits LED current, while connector J2 provides access to the +5 V output and J3 provides the ground connection.

#### 3.3 Filtering and Output Indication

U1 is the LM7805 three-terminal positive linear voltage regulator. Its input pin receives the protected DC supply, the ground pin is connected to the common return, and the output pin provides the regulated +5 V rail. Capacitor C1, rated 0.33 uF in the design, is connected near the input side to provide local bypassing.

#### 3.2 Voltage Regulation Stage

The nominal +12 V DC input enters through connector J1. Diode D1 (1N4007) is placed in series with the input path and provides simple protection against accidental reverse-polarity connection. Under correct polarity, the diode conducts and supplies the regulator input stage.

#### 3.1 Input Protection Stage

The +12 V input is applied through connector J1 and passes through diode D1 (1N4007). Capacitor C1 (0.33 uF) is connected at the input side of the LM7805 regulator. U1 regulates the input to approximately +5 V. Capacitor C2 (0.1 uF) is connected at the output side for stability and filtering. The regulated output is available at J2. LED D2 and resistor R1 (330 ohm) form the output indication branch, while J3 provides the ground connection.

## Chapter 4

### Schematic Design in eSim

Before PCB generation, the electrical connectivity was checked in the schematic and the design references were kept consistent. This is important because the PCB netlist and pad connectivity are derived from the schematic relationships.

The schematic was organized so that the signal and power flow proceeds from left to right: +12 V input, protection diode, input capacitor, LM7805 regulator, output capacitor, +5 V connector, and LED indicator. A common ground net links the regulator, capacitors, output return, and indicator branch. This arrangement makes the circuit easier to inspect and reduces ambiguity during transfer to the PCB editor.

The schematic was created in the eSim schematic editor. Electrical connections were completed before proceeding to footprint assignment and PCB generation.

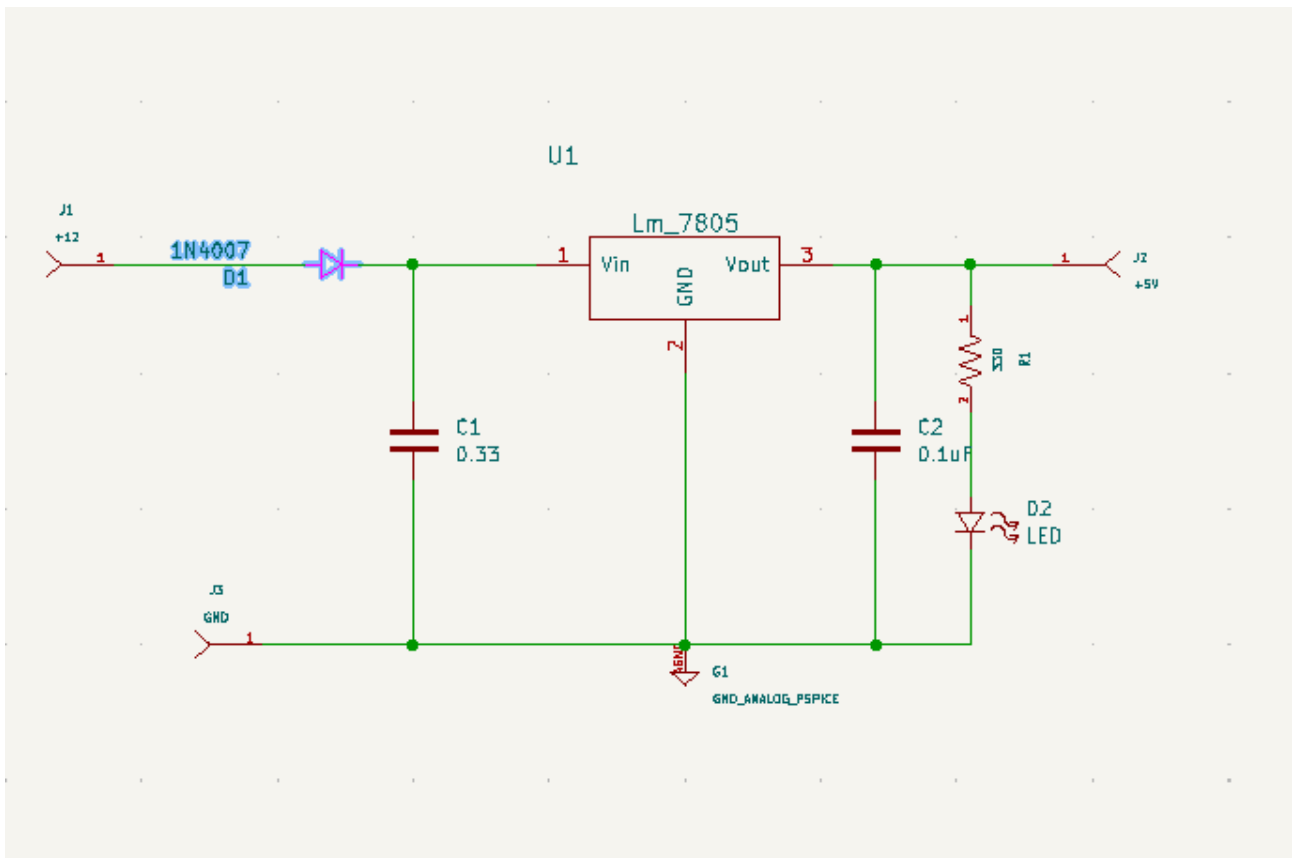


Figure 1: Schematic of the 12 V to 5 V LM7805 regulated power supply.

## Chapter 5

### Footprint Selection and Assignment

Correct footprint assignment is critical because a schematic symbol describes electrical behavior, whereas the footprint defines the physical pad geometry, hole spacing, package outline, and assembly orientation. A mismatch can produce a PCB that is electrically correct in software but impossible or inconvenient to assemble.

Through-hole footprints were selected for the components to make the design easy to inspect and suitable for manual assembly. The LM7805 uses a vertical TO-220-3 footprint; the 1N4007 uses a horizontal DO-41 footprint; C1 and C2 use disc-capacitor footprints; the LED uses a 5 mm through-hole footprint; R1 uses an axial resistor footprint; and J1, J2, and J3 use single-pin 2.54 mm vertical headers.

Appropriate through-hole footprints were assigned according to the physical package and pin configuration of each component. The selected footprints are summarized below.

| Reference | Component | Assigned footprint                                             |
|-----------|-----------|----------------------------------------------------------------|
| C1        | 0.33 uF   | Capacitor THT:C Disc D5.1mm W3.2mm P5.00mm                     |
| C2        | 0.1 uF    | Capacitor THT:C Disc D5.1mm W3.2mm P5.00mm                     |
| D1        | 1N4007    | Diode THT:D DO-41 SOD81 P10.16mm Horizontal                    |
| D2        | LED       | LED THT:LED D5.0mm                                             |
| J1        | +12 V     | Connector PinHeader 2.54mm:PinHeader 1x01 P2.54mm Vertical     |
| J2        | +5 V      | Connector PinHeader 2.54mm:PinHeader 1x01 P2.54mm Vertical     |
| J3        | GND       | Connector PinHeader 2.54mm:PinHeader 1x01 P2.54mm Vertical     |
| R1        | 330 ohm   | Resistor THT:R Axial DIN0207 L6.3mm D2.5mm P10.16mm Horizontal |
| U1        | LM7805    | Package TO SOT THT:TO-220-3 Vertical                           |

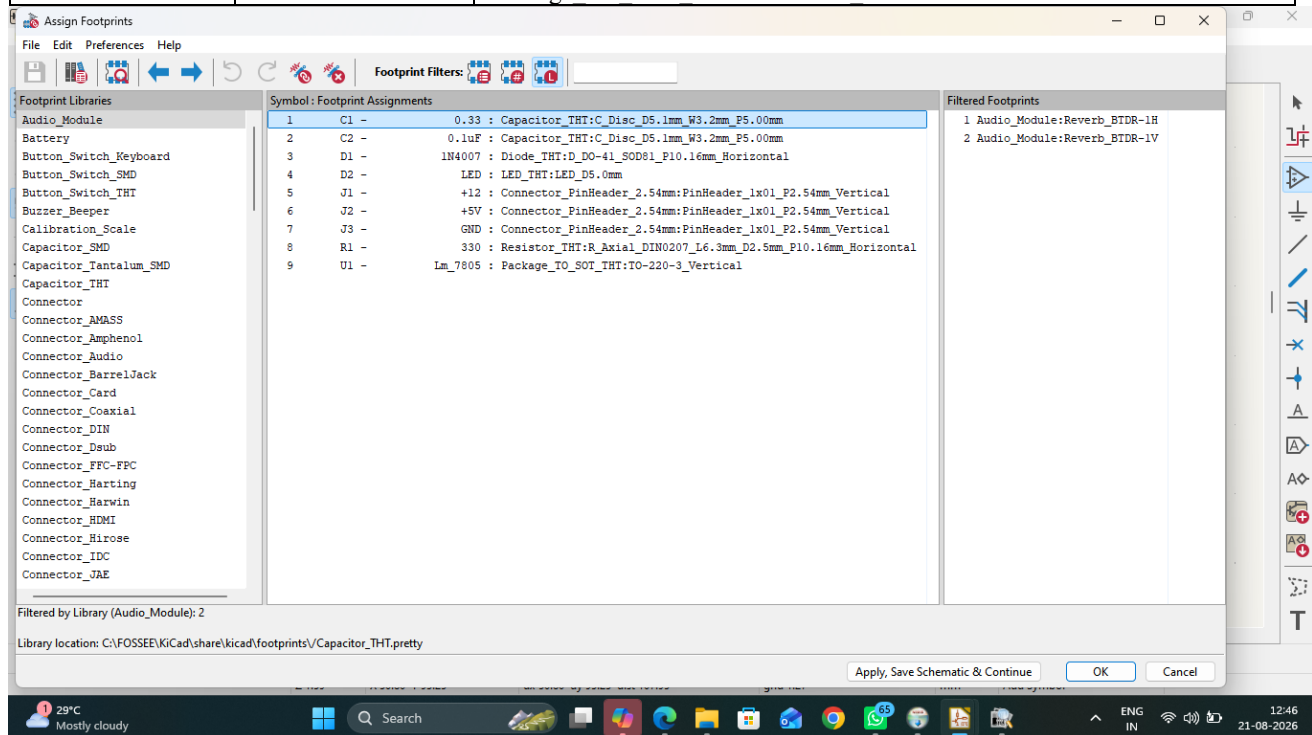


Figure 2: Footprint assignment window showing the selected component footprints.

## Chapter 6

### PCB Layout and Component Placement

The rectangular board boundary was created on the Edge.Cuts layer. Components were kept within the outline and sufficient separation was maintained so that pads, outlines, and routed tracks remained readable. The regulator was placed where its three terminals could be routed directly to input, ground, and output networks.

Component placement was performed with functional grouping in mind. The input connector and protection diode were positioned on the left side, followed by the input capacitor and regulator. The output capacitor and +5 V connector were positioned toward the right, while the LED and resistor were placed in the output section. This produces a visible left-to-right power path and avoids unnecessary track length.

After updating the PCB from the schematic, the footprints were placed inside the board outline. The input components were arranged toward the left, the regulator and filtering components near the center, and the output/indicator components toward the right. This arrangement follows the signal and power flow and keeps the layout easy to inspect.

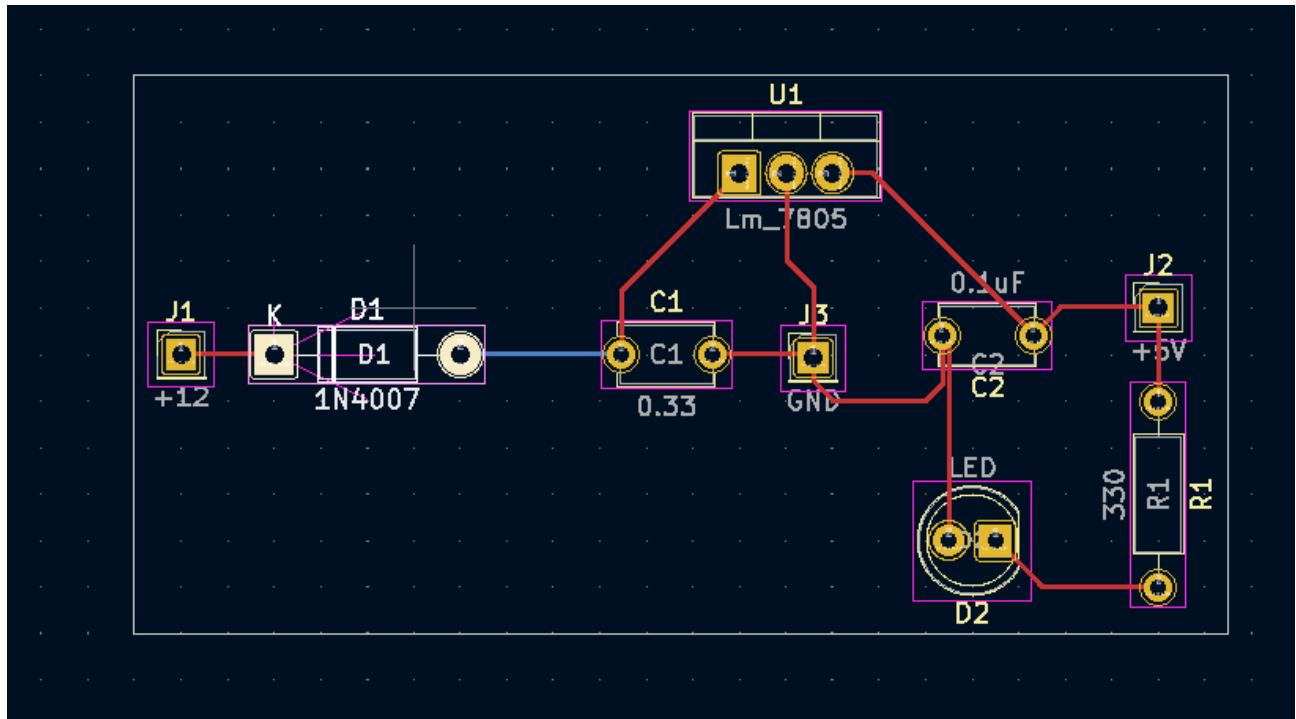


Figure 3: PCB component placement and routed connections.

## Chapter 7

### Routing Strategy and Two-Layer Design

During routing, the goal was not only to remove all unrouted connections but also to keep the board visually organized. Power and ground connections were kept short where practical, and the capacitors were positioned near the regulator terminals so that their associated tracks did not need to travel across the board.

The board was implemented as a two-copper-layer design using F.Cu and B.Cu. Routing was performed according to the schematic net connectivity, with direct paths preferred for the input and regulated-output rails. The use of two layers provides flexibility in resolving crossings and demonstrates compliance with the Task 7 requirement for at least two copper layers.

Tracks were routed between the pads according to the schematic netlist. The routing was kept direct and organized, with the input, regulator, ground, output capacitor, output connector, and LED indicator connections completed without leaving unrouted nets. The board outline was created on the Edge.Cuts layer.

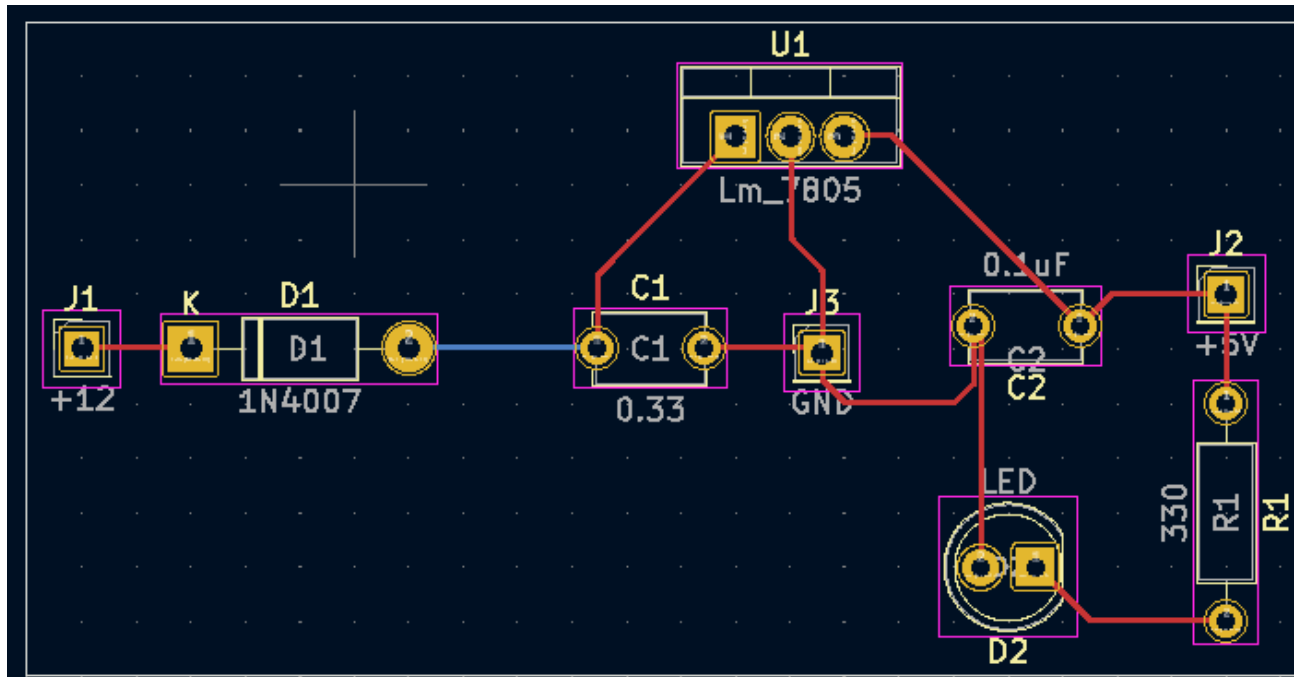


Figure 4: Final routed PCB layout.

## Chapter 8

### Design Rule Check and Verification

A clean DRC result indicates that the board satisfies the configured design rules, but it does not replace engineering review. Component orientation, connector labeling, physical accessibility, thermal performance, and the suitability of footprints must still be considered separately.

Design Rule Check is the final automated verification step used to detect common PCB-layout problems such as insufficient clearance, unconnected items, overlapping features, and rule violations. Earlier clearance issues were corrected before the final check. The final DRC window showed zero violations and zero unconnected items, with zero errors and zero warnings.

The PCB was verified using the Design Rule Checker. The final DRC result shows zero violations and zero unconnected items, with zero errors and zero warnings. This confirms that the routed PCB satisfies the configured design rules and that all required nets are connected.

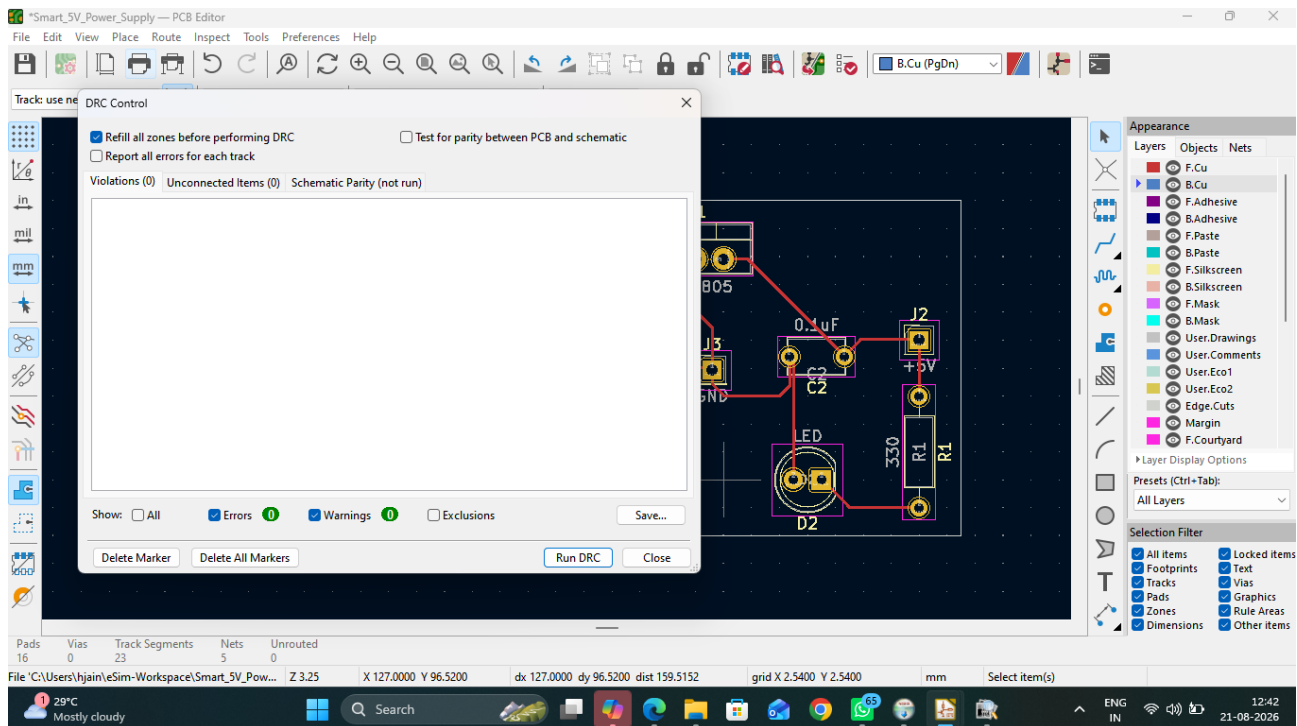


Figure 5: Final DRC result showing 0 violations, 0 unconnected items, 0 errors and 0 warnings.

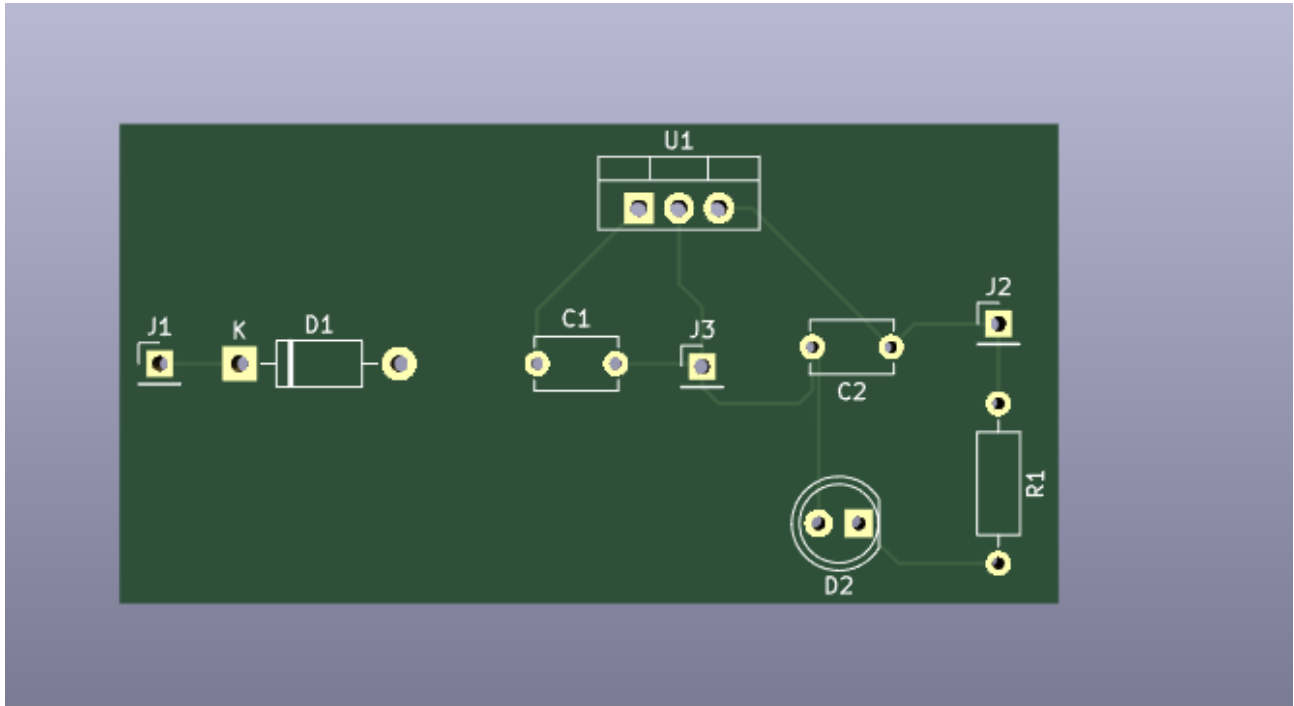
## Chapter 9

### 3D Visualization

The visualization in this report should be interpreted as a design-review representation rather than a photograph of a fabricated board. The final physical appearance depends on the actual manufactured PCB, component packages, soldering, and assembly process.

The 3D Viewer was used as a visual verification tool after routing. It helps confirm the board outline and the relative physical positions of the assigned component packages. The 3D view is especially useful for identifying obvious placement or orientation problems that may not be as apparent in the two-dimensional PCB editor.

The completed PCB was opened in the 3D Viewer to inspect the physical arrangement of the board and component footprints. The 3D view provides a visual check of component positions and the overall board form.



*Figure 6: 3D visualization of the designed PCB.*

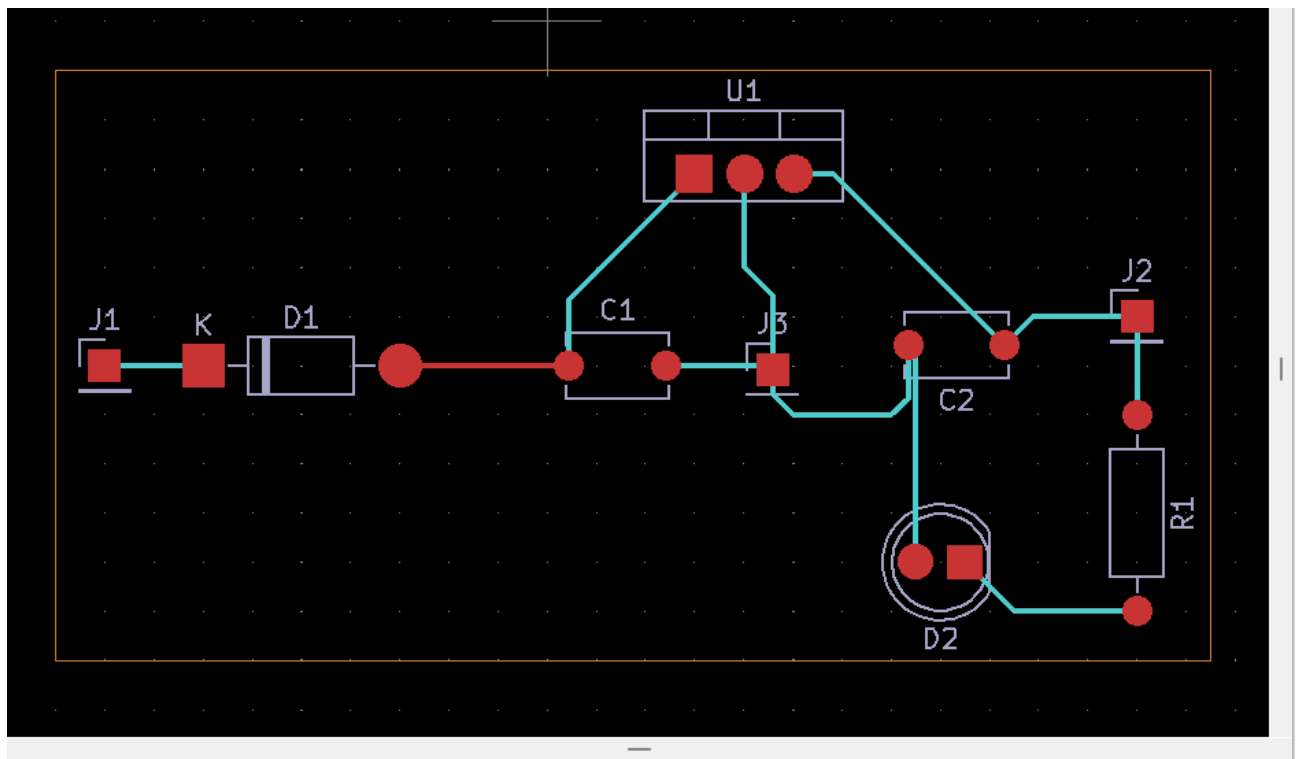
## Chapter 10

### Gerber and Drill File Generation

The generated files were opened in the Gerber Viewer to confirm that the board outline, pads, copper tracks, masks, and fabrication layers were present. This verification step is important because manufacturing files are the final data delivered to a PCB fabricator; therefore, they should be inspected independently of the editable PCB project.

After successful DRC, Gerber outputs were generated for the copper, solder-mask, silkscreen, and Edge.Cuts information required for PCB fabrication. Separate plated-through-hole and non-plated-through-hole drill files were also generated. The Gerber job file groups the layer information for manufacturing review.

Gerber manufacturing files and drill files were generated after successful DRC. The generated layers were opened in the Gerber Viewer to verify the board outline, copper data, solder mask/silkscreen information, and drill locations before submission or fabrication.



*Figure 7: Gerber Viewer showing the generated PCB fabrication data.*

## Chapter 11

### Design Considerations and Engineering Decisions

For a production-oriented version, additional considerations would include connector robustness, input fuse or resettable protection, transient protection, thermal analysis of the LM7805, copper width based on expected load current, mounting holes, polarity markings, test points, and a more explicit ground strategy. These additions were not required to demonstrate the core Task 7 PCB workflow.

The design emphasizes clarity and manufacturability for an internship screening task. Through-hole parts simplify visual inspection and manual assembly. The power path is arranged in a logical sequence, the board outline is simple, and the input/output connectors are located near the board edges for accessibility.

- The schematic was kept simple and organized to make the power path easy to understand.
- Through-hole footprints were selected for straightforward assembly and demonstration.
- Input and output capacitors were placed close to the LM7805 connections in the layout.
- The PCB outline was defined using the Edge.Cuts layer.
- Both front and back copper layers are included in the PCB/Gerber layer set as required for a two-layer board.
- DRC was completed before final manufacturing-file generation.

## Chapter 12

### Results and Discussion

The project therefore meets its educational objective of demonstrating the transition from a circuit schematic to a verified two-layer PCB design. Actual electrical performance such as output regulation under varying load, ripple, temperature rise, and maximum usable current would require fabrication and laboratory testing and is not claimed from the software design alone.

The most important verification result is the final DRC status of 0 errors, 0 warnings, 0 violations, and 0 unconnected items. The manufacturing outputs were also generated and visually inspected in the Gerber Viewer. These results demonstrate successful completion of the software-based PCB design and fabrication-output workflow required for the screening task.

The complete editable eSim/KiCad project was produced together with the routed PCB, 3D visualization, DRC evidence, Gerber layers, drill files, and technical report. The final design contains the intended +12 V input, LM7805 regulation stage, +5 V output, input/output filtering, and LED indication network.

A functional 12 V to 5 V regulated power-supply PCB design was completed through the required eSim/KiCad workflow. The schematic, footprints, PCB layout, routed connections, board outline, DRC verification, 3D visualization, Gerber files and drill files were produced successfully. The final DRC screenshot confirms zero violations and zero unconnected items.

## Chapter 13

### Limitations and Scope for Improvement

The present work is a PCB-design and manufacturing-output demonstration. It does not include a fabricated prototype or measured laboratory data. Consequently, the report does not claim measured load regulation, thermal performance, ripple, efficiency, or long-term reliability.

The design can be extended by adding a power switch, fuse or resettable protection, reverse-polarity protection with lower forward loss, input transient suppression, mounting holes, clearly marked test points, and wider power tracks for a defined current rating. A heatsink provision for the LM7805 may also be considered when the expected dissipation is significant.

A future version may include a larger input bulk capacitor when the source is not already well filtered, a ground copper zone, improved silkscreen annotations, and laboratory validation using a variable load. Such testing would allow comparison of theoretical operation with measured output voltage, regulator temperature, and transient response.

## Chapter 14

### Conclusion

The resulting files form a complete software-based submission for the screening task. Further development can focus on physical fabrication and laboratory testing to evaluate electrical and thermal performance under real operating conditions.

The project achieved a clean final DRC result and produced the Gerber and drill outputs required for manufacturing review. Beyond the specific power-supply circuit, the activity provided practical experience in organizing a PCB project, selecting physical packages, translating schematic connectivity into copper routing, and verifying a design before fabrication.

The Smart 5V Power Supply project demonstrates the complete PCB-design workflow expected in Screening Task 7 of the eSim Semester Long Internship - Autumn 2026. Starting from the LM7805-based schematic, the work progressed through footprint assignment, board generation, component placement, two-layer routing, DRC verification, 3D visualization, and fabrication-file generation.

The project demonstrates the complete PCB design flow from schematic capture to fabrication-output verification. The LM7805-based circuit provides a simple example for understanding footprint mapping, component placement, PCB routing, design-rule checking, 3D visualization and Gerber generation in eSim/KiCad. The design is ready to be packaged with the complete eSim project folder and supporting documentation for submission.

## Chapter 15

### Final Submission Files

- Complete eSim project folder, including .kicad\_sch, .kicad\_pcb, .kicad\_pro and any required project/library files.
- This detailed PDF report.
- Generated Gerber files and PTH/NPTH drill files.
- PCB design proposal.
- GitHub repository link and Pull Request link, as required by the Task 7 submission instructions.